



The Scots College

2002

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.
- Answer each question in a SEPARATE writing booklet.
- Extra writing booklets are available.
- Question Papers are to be handed in.

Total marks – 84

- Attempt Questions 1–7.
- All questions are of equal value.

Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)

Marks

- a. Differentiate $4x \sin^{-1} x$ **2**
- b. If $\int_{-a}^a \frac{dx}{1+x^2} = \frac{\pi}{2}$ Find the value of a . **2**
- c. Find the coordinates of the point P that divides the interval joining $(-4, 3)$ and $(2, -7)$ externally in the ratio 4:3. **2**
- d. If $\log_a b = 2.8$ and $\log_a c = 4.1$, find $\log_a bc$. **1**
- e. Solve for x : $\frac{x+2}{x} \leq 3, x \neq 0$. **2**
- f. Evaluate $\int_{0.5}^1 4x(2x-1)^5 dx$ by making the substitution $u = 2x-1$. **3**

End of Question 1

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Taking $x = 0.5$ as a first approximation for the root of $\log_e x = -x$, use Newton's method to find a second approximation. (Answer correct to 3 significant figures.) **3**

- b. Prove that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ **4**

- c. Find the size of α and β in the following diagram (giving reasons). **2**

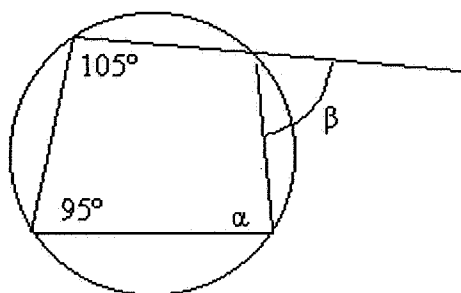


Diagram NOT to scale.

- d.

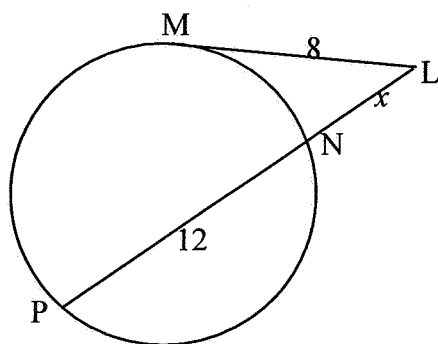


Diagram NOT to scale.

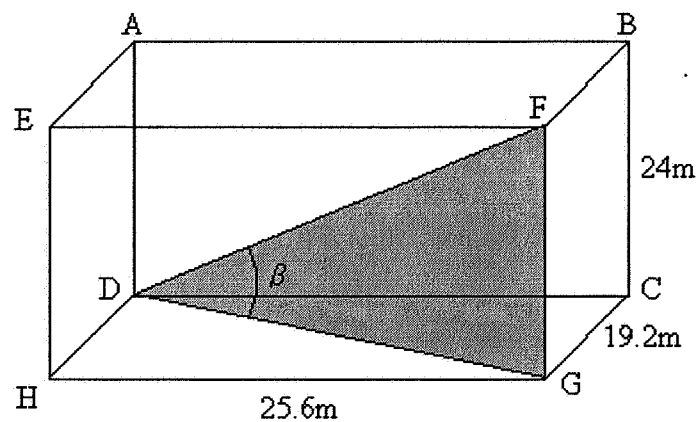
- LM is a tangent to the circle, while LNP is a secant intersecting the circle at N and P. Given that $LM = 8$, $NP = 12$ and $LN = x$, find x . **3**

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Find, for $0 \leq x \leq 2\pi$, all solutions for the equation $\cos x = \sin 2x$. **3**
- b. If α, β and γ are the roots of the cubic polynomial equation $x^3 + 8x^2 - 4x - 6 = 0$ **3**
Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$.
- c. Find the term independent of x in the expansion of $(3x^4 + \frac{1}{x})^{10}$ **3**
- d. Calculate the value of β in the following rectangular prism. **3**
(Answer to the nearest minute.)

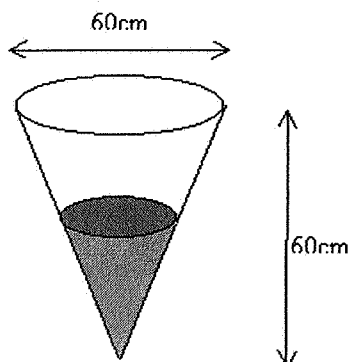


End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Sand is being poured into a conical container at a constant rate of $36\text{cm}^3\text{s}^{-1}$. The height and diameter of the container are both 60cm. After t seconds the depth of the sand in the container is h cm.



- | | | |
|------|---|---|
| i. | Show that the depth of sand in the container after 5 seconds is 8.826cm correct to 2d.p. | 2 |
| ii. | Find the rate at which the depth of sand is changing after 5 seconds. (answer to 3d.p.) | 2 |
| iii. | Find the rate at which the surface area, S , of the sand in the container is changing when the depth of the sand is 20cm. | 2 |

- b. i. Find the domain and range of the function $y = 4 \cos^{-1}\left(\frac{x}{3}\right)$. 1
- ii. Sketch the graph of the function $y = 4 \cos^{-1}\left(\frac{x}{3}\right)$ showing clearly the intercepts on the coordinate axes and the coordinates of any endpoints. 2
- iii. Find the area of the region in the first quadrant bounded by the curve $y = 4 \cos^{-1}\left(\frac{x}{3}\right)$ and the coordinate axes 3

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. The parabola given by $x = 2at$ and $y = at^2$ has points P and Q where $t = p$ and $t = q$ respectively.
Prove:

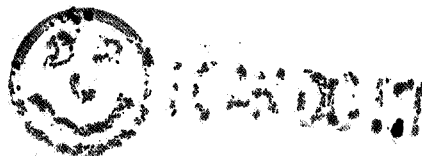
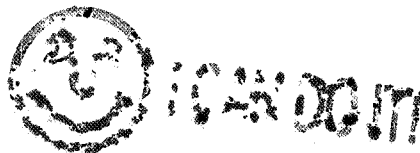
- i. The equation of the chord PQ is given by $y - \frac{(p+q)x}{2} + apq = 0$ 2
- ii. The equation of the tangent at P is given by $y - px + ap^2 = 0$ 2
- iii. The tangents at the ends of any focal chord meet on the directrix and are perpendicular to each other. 3

- b. At time t the temperature T° Celsius of a piece of iron in a room of constant temperature 30° Celsius, is decreasing according to the equation

$$\frac{dT}{dt} = -k(T - 30) \text{ for some constant } k > 0.$$

- i. Verify that $T = 30 + Ae^{-kt}$, A constant, is a solution of the equation. 2
- ii. The initial temperature, T , of the piece of iron is 100°C and it falls to 60°C after 20 minutes. Find the temperature of the body after a further 10 minutes. (Answer to nearest degree.) 3

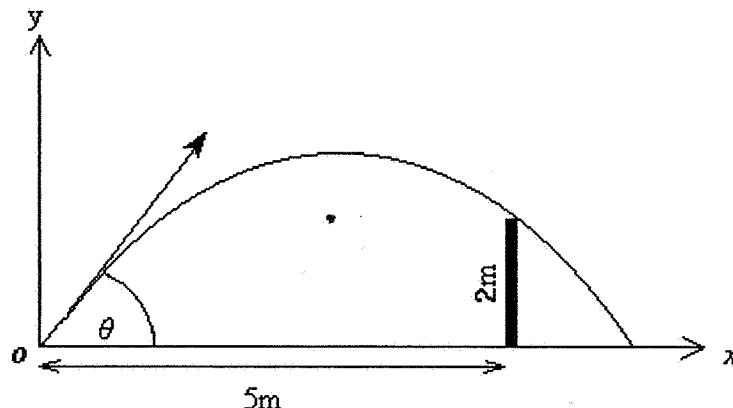
End of Question 5



Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. A ball is shot out of a small cannon on the ground at point O with velocity 15ms^{-1} , fired at an angle of θ with the horizontal.



- i. Taking acceleration due to gravity to be a constant 10ms^{-2} , show that the equation for the horizontal (x) and vertical (y) components of the particles displacement from the origin O are given by $x = 15t \cos \theta$ and $y = -5t^2 + 15t \sin \theta$ 2
- ii. Show that the Cartesian equation for displacement is given by $y = \frac{-x^2}{45} \sec^2 \theta + x \tan \theta$ 2
- iii. The ball just clears a 2 metre high fence that is 5 metres from the origin. Find two values of θ (measured in degrees) for this to happen. (Give your answer to the nearest minute) 3

- b. A particle is moving such that its acceleration is given by $\ddot{x} = -16x$. The particle has an initial displacement of 3m and an initial velocity of 12ms^{-1} .

- i. Show the velocity is given by $\dot{x} = 4\sqrt{18 - x^2}$ 3
- ii. Find the equation for the particle displacement, x , over time t . 2

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Use the principal of mathematical induction to prove that, for every positive integer n , $13 \times 6^n + 2$ is divisible by 5. 4
- b. A particle moves in a straight line. Its displacement x metres from the origin, after t seconds is given by $x = \sin^2 5t + 2$, $t > 0$.
- i. Find the time when the particle is first at $x = \frac{5}{2}$. 2
- ii. In what direction is the particle travelling when it is first at $x = \frac{5}{2}$? 1
- iii. Express the acceleration of the particle in terms of x . 2
- iv. Hence, or otherwise, show that the particle is undergoing simple harmonic motion. 2
- v. State the period of the motion. 1

End of paper

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Use the principal of mathematical induction to prove that, for every positive integer n , $13 \times 6^n + 2$ is divisible by 5. 4
- b. A particle moves in a straight line. Its displacement x metres from the origin, after t seconds is given by $x = \sin^2 5t + 2$, $t > 0$.
- i. Find the time when the particle is first at $x = \frac{5}{2}$. 2
- ii. In what direction is the particle travelling when it is first at $x = \frac{5}{2}$? 1
- iii. Express the acceleration of the particle in terms of x . 2
- iv. Hence, or otherwise, show that the particle is undergoing simple harmonic motion. 2
- v. State the period of the motion. 1

End of paper

QUESTION ONE SOLUTIONS

$$\begin{aligned} \text{a. } \frac{d}{dx} 4x \sin^{-1} x &= \sin^{-1} x + 4x \times \frac{1}{\sqrt{1-x^2}} \\ &= 4 \sin^{-1} x + \frac{4x}{\sqrt{1-x^2}} \end{aligned}$$

$$\begin{aligned} \text{b. } \int_{-a}^a \frac{dx}{1+x^2} &= [\tan^{-1} x]_{-a}^a \\ &= \tan^{-1} a - \tan^{-1}(-a) \\ \text{so } \tan^{-1} a - \tan^{-1}(-a) &= \frac{\pi}{2} \\ 2 \tan^{-1} a &= \frac{\pi}{2} \\ \tan^{-1} a &= \frac{\pi}{4} \therefore a = 1 \end{aligned}$$

$$\begin{aligned} \text{c. } x_1 &= -4, y_1 = 3 \\ x_2 &= 2, y_2 = -7 \\ m &= 4, n = -3 \\ P &= \left(\frac{4 \cdot 2 + (-3) \cdot (-4)}{4 + (-3)}, \frac{4 \cdot (-7) + (-3) \cdot 3}{4 + (-3)} \right) \\ P &= (20, -37) \end{aligned}$$

$$\begin{aligned} \text{d. } \log_a bc &= \log_a b + \log_a c \\ &= 2.8 + 4.1 \\ &= 6.9 \end{aligned}$$

QUESTION TWO SOLUTIONS

$$\text{a. let } f(x) = \log_e x + x$$

$$\text{then } f'(x) = \frac{1}{x} + 1$$

$$\text{let } x_1 = 0.5$$

$$\begin{aligned} \text{now } x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ x_2 &= 0.5 - \frac{f(0.5)}{f'(0.5)} \\ x_2 &= 0.5 - \frac{\ln(0.5) + 0.5}{3} \\ x_2 &= 0.564 \text{ (to 3 Sig. figs)} \end{aligned}$$

$$\begin{aligned} \text{c. } \frac{x+2}{x} &\leq 3 \text{ mult. b.s. by } x^2 \\ x(x+2) &\leq 3x^2 \\ 0 &\leq 2x^2 - 2x \\ 0 &\leq 2x(x-1) \\ x &< 0 \text{ and } x \geq 1 \text{ (note } x \neq 0) \end{aligned}$$

$$\begin{aligned} \text{f) let } u &= 2x - 1 \text{ so} \\ du &= 2dx \text{ and } 2x = u + 1 \\ \text{when } x &= 0.5, u = 0 \text{ and when } x = 1, u = 1 \end{aligned}$$

$$\begin{aligned} \text{so } \int_{0.5}^1 4x(2x-1)^5 dx &\text{ becomes} \\ &= \int_{0.5}^1 2x(2x-1)^5 2dx \\ &= \int_{0.5}^1 2x(2x-1)^5 2dx \\ &= \int_0^1 (u+1)(u)^5 du \\ &= \int_0^1 (u^6 + u^5) du \\ &= \left[\frac{u^7}{7} + \frac{u^6}{6} \right]_0^1 \\ &= \frac{13}{42} \end{aligned}$$

$$\text{b. Prove that } \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\begin{aligned} \text{L.H.S.} &= \cos(2\theta + \theta) \\ &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= \cos \theta (2 \cos^2 \theta - 1) - \sin \theta (2 \sin \theta \cos \theta) \\ &= 2 \cos^3 \theta - \cos \theta - 2 \sin^2 \theta \cos \theta \\ &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta (1 - \cos^2 \theta) \\ &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta + 2 \cos^3 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta \text{ (=RHS)} \end{aligned}$$

c. $\alpha = 75^\circ$ (opposite angles in a cyclic quad are supp.) and $\beta = 95^\circ$ (Exterior angle equal to opp. int. angle)

d. $LM^2 = LN \cdot LP$
 $8^2 = x(12+x)$
 $64 = 12x + x^2$
 $x^2 + 12x - 64 = 0$

$(x+16)(x-4) = 0$
 so $x = -16$ or $x = 4$
 since $x > 0$ the only solution will be $x = 4$

QUESTION 3 SOLUTIONS

a.

$\cos x = \sin 2x$
 so $0 = \sin 2x - \cos x$
 $0 = 2 \sin x \cos x - \cos x$
 $0 = \cos x(2 \sin x - 1)$
 $\therefore 2 \sin x - 1 = 0$
 and $\cos x = 0$

now $0 \leq x \leq 2\pi$
 so for $2 \sin x - 1 = 0$

$\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

and for $\cos x = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2}$

So there are 4 solutions

$x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b.

$\alpha + \beta + \gamma = \frac{-b}{a} \quad \alpha\beta + \alpha\gamma + \gamma\beta = \frac{c}{a} \quad \alpha\beta\gamma = \frac{-d}{a}$
 $\frac{-8}{1} \quad \frac{-4}{1} \quad \frac{-6}{1}$
 $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$
 $\frac{4}{3}$

c.i. expansion of $(3x^4 + \frac{1}{2x})^{10}$

$(a+b)^n = \sum_{k=0}^n {}^nC_k a^{n-k} b^k$

$T_{k+1} = {}^{10}C_k (3x^4)^{10-k} (\frac{1}{2x})^k$

$T_{k+1} = {}^{10}C_k 3^{10-k} \times x^{40-4k} \times x^{-k}$

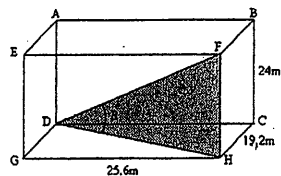
$T_{k+1} = {}^{10}C_k 3^{10-k} \times x^{40-5k}$

now $40 - 5k = 0$ so $k = 8$
 So term independent of x will be

$= {}^{10}C_8 (3x^4)^{10-8} (\frac{1}{2x})^8$

$= {}^{10}C_8 3^2$

$= 405$



d. $DH^2 = 25.6^2 + 19.2^2$
 $DH^2 = 25.6^2 + 19.2^2$
 $DH = 32$

$\therefore \tan \beta = \frac{24}{32}$

$\therefore \beta = 36^\circ 52'$

QUESTION FOUR SOLUTIONS

i. Now Volume of Sand in container is

Now $\frac{dV}{dt} = 36$ so after 5 seconds

$V = 180 \text{ cm}^3$ will have poured in.

Note: $h = d$ ($= 2r$) so $r = h/2$

$V = \frac{1}{3} \pi r^2 h \rightarrow V = \frac{1}{3} \pi (\frac{h}{2})^2 h$

$V = \frac{\pi h^3}{12} \quad \therefore \frac{dV}{dh} = \frac{\pi h^2}{4}$

$\therefore 180 = \frac{\pi h^3}{12}$

$h = \sqrt[3]{\frac{12 \times 180}{\pi}}$

$h = 8.826 \text{ cm}$

ii. We are after $\frac{dh}{dt}$

and we know $\frac{dV}{dt} = 36$

now $\frac{dh}{dt} = \frac{dV}{dt} \cdot \frac{dh}{dV}$

$\frac{dh}{dt} = 36 \cdot \frac{4}{\pi h^2}$

when $h = 8.826 \text{ cm}$

$\frac{dh}{dt} = 36 \cdot \frac{4}{\pi (8.826)^2}$

$\frac{dh}{dt} = 0.588 \text{ cms}^{-1}$

iii. We are after $\frac{dS}{dt}$

now S.A. of the sand $\dot{S} = \pi r^2$

$r = h/2$ so $S = \pi r^2$

$S = \pi (\frac{h}{2})^2$

$S = \frac{\pi h^2}{4}$

so $\frac{dS}{dh} = \frac{\pi h}{2}$

NOW $\frac{dS}{dt} = \frac{dV}{dt} \cdot \frac{dS}{dV}$

$\frac{dS}{dt} = \frac{dV}{dt} \cdot \frac{dS}{dh} \cdot \frac{dh}{dV}$

$\frac{dS}{dt} = 36 \cdot \frac{\pi h}{2} \cdot \frac{4}{\pi h^2}$

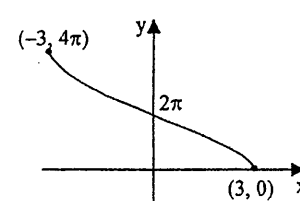
$\frac{dS}{dt} = \frac{72}{h}$

$\frac{dS}{dt} = \frac{72}{20}$

$\frac{dS}{dt} = 3.6 \text{ cm}^2 \text{ s}^{-1}$

b) i. The domain $-3 \leq x \leq 3$, The range $0 \leq y \leq 4\pi$

ii.



iii. $A = \int_0^{2\pi} 3 \cos \frac{y}{4} dy$

$A = 3 \left[4 \sin \frac{y}{4} \right]_0^{2\pi}$

$A = 3 \left[4 \sin \frac{\pi}{2} - 4 \sin 0 \right]$

$A = 12 \text{ units}^2$

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QUESTION FIVE SOLUTIONS

a.

$P(2ap, ap^2)$ and $Q(2aq, aq^2)$

Gradient of Chord PQ

$$\begin{aligned}\text{Gradient} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{a(p^2 - q^2)}{2a(p - q)} \\ &= \frac{a(p - q)(p + q)}{2a(p - q)} \\ &= \frac{p + q}{2}\end{aligned}$$

Eqn of chord

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = \frac{p + q}{2}(x - 2ap)$$

$$y - ap^2 = \frac{(p + q)x}{2} - \frac{2ap^2}{2} - \frac{2apq}{2}$$

$$y - ap^2 = \frac{(p + q)x}{2} - ap^2 - apq$$

$$y - \frac{(p + q)x}{2} + apq = 0$$

b)i) If $T = 30 + Ae^{-kt}$ is a solution then

$$\frac{dT}{dt} = -k(T - 30)$$

$$\text{so } \frac{d(30 + Ae^{-kt})}{dt} = -k(30 + Ae^{-kt} - 30)$$

$$-kAe^{-kt} = -kAe^{-kt} \text{ so true}$$

ii) Eqn of tangent

$$x^2 = 4ay \rightarrow y = \frac{x^2}{4a} \rightarrow \frac{dy}{dx} = \frac{x}{2a}$$

when $x = 2ap$ gradient $= p$

Eqn of tangent at P

$$y - ap^2 = px - 2ap^2$$

$$\therefore y - px + ap^2 = 0$$

iii. If PQ is a focal chord then $pq = -1$

$$\text{Tng at P } y - px + ap^2 = 0 \quad (1)$$

$$\text{Tng at Q } y - qx + aq^2 = 0 \quad (2)$$

mult (1) by q and (2) by p

$$yq - pqx + ap^2q = 0 \quad (3)$$

$$yp - pqx + aq^2q = 0 \quad (4)$$

$$(4) - (3)$$

$$y(p - q) - apq(p - q) = 0$$

$$(p - q)(y - apq) = 0$$

$$\therefore y = apq \text{ now } pq = -1 \text{ so } y = -a$$

this means all y values are on the directrix

Gradient of tangent through P is p

Gradient of tangent through Q is q

now $pq = -1$ so tangents are perpendicular.

ii) When $t = 0, T = 100$

$$\therefore 100 = 30 + Ae^0$$

$$\therefore A = 70 \text{ When } t = 20, T = 60$$

$$\text{so } 60 = 30 + 70e^{-20k}$$

$$e^{-20k} = \frac{30}{70}$$

$$k = -\ln\left(\frac{3}{7}\right)/20$$

so now after a further 10 minutes $t = 30$

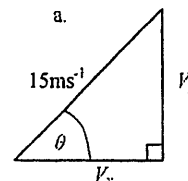
$$T = 30 + 70e^{-30\ln(3/7)/20}$$

$$T = 49.6396$$

$$T = 50^\circ \text{ to nearest degree.}$$

Question 6 SOLUTIONS

a.



Horizontal Velocity

$$\frac{V_x}{15} = \cos \theta$$

$$V_x = 15 \cos \theta$$

now displacement $s = ut + 0.5at$

horizontally $a = 0$

$$\text{so } x = (15 \cos \theta)t + 0$$

$$x = 15t \cos \theta$$

Vertical Velocity

$$\frac{V_y}{15} = \sin \theta$$

$$V_y = 15 \sin \theta$$

vertically $a = -10$

$$y = (15 \sin \theta)t + 0.5 \times -10 \times t^2$$

$$y = -5t^2 + 15t \sin \theta$$

$$\text{ii. From } x = 15t \cos \theta, t = \frac{x}{15 \cos \theta}$$

$$\text{subst. into } y = -5t^2 + 15t \sin \theta$$

$$y = -5\left(\frac{x}{15 \cos \theta}\right)^2 + 15\left(\frac{x}{15 \cos \theta}\right) \sin \theta$$

$$y = \frac{-5x^2}{225 \cos^2 \theta} + \frac{15x \sin \theta}{15 \cos \theta}$$

$$y = \frac{-x^2}{45} \sec^2 \theta + x \tan \theta$$

$$\text{iii. now when } x = 5, y = 2$$

$$2 = \frac{-25}{45} \sec^2 \theta + 5 \tan \theta$$

$$18 = -5 \sec^2 \theta + 45 \tan \theta$$

$$18 = -5(1 - \tan^2 \theta) + 45 \tan \theta$$

$$5 \tan^2 \theta - 45 \tan \theta + 23 = 0$$

$$\text{let } x = \tan \theta \text{ to get } 5x^2 - 45x + 23 = 0$$

$$x = \frac{45 \pm \sqrt{(-45)^2 - 4 \times 5 \times 23}}{10}$$

$$\tan \theta = \frac{45 \pm \sqrt{1565}}{10}$$

$$\theta = \tan^{-1}\left(\frac{45 + \sqrt{1565}}{10}\right)$$

$$\theta = 83^\circ 15' \text{ (to nearest minute)}$$

$$\theta = \tan^{-1}\left(\frac{45 - \sqrt{1565}}{10}\right)$$

$$\theta = 28^\circ 33' \text{ (to nearest minute)}$$

b.

$$\ddot{x} = -16x$$

$$\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$\text{so } \frac{1}{2}v^2 = \int \ddot{x} dx$$

$$\text{so } \frac{1}{2}v^2 = \int -16x dx$$

$$\frac{1}{2}v^2 = -8x^2 + c$$

$$\text{when } x = 3, v = 12$$

$$\text{so } \frac{1}{2} \times 144 = -72 + c$$

$$c = 144$$

$$\frac{1}{2}v^2 = -8x^2 + 144$$

$$v = \pm \sqrt{-16x^2 + 288}$$

$$v = \pm \sqrt{-16x^2 + 288}$$

$$v = \pm 4\sqrt{18 - x^2}$$

$$\text{so } \frac{dx}{dt} = \pm 4\sqrt{18 - x^2}$$

$$\text{so } \frac{dt}{dx} = \frac{1}{\pm 4\sqrt{18 - x^2}}$$

$$\text{and } \pm t = -\frac{1}{4} \cos^{-1} \frac{x}{3\sqrt{2}} + c$$

$$\text{when } t = 0, x = 3$$

$$\therefore 0 = -\frac{1}{4} \cos^{-1} \frac{3}{3\sqrt{2}} + c$$

$$\therefore c = \frac{\pi}{16}$$

$$\pm t = -\frac{1}{4} \cos^{-1} \frac{x}{3\sqrt{2}} + \frac{\pi}{16}$$

$$\pm 4t = -\cos^{-1} \frac{x}{3\sqrt{2}} + \frac{\pi}{4}$$

$$x = 3\sqrt{2} \left[\cos\left(\frac{\pi}{4} \pm 4t\right) \right]$$

QUESTION SEVEN SOLUTIONS.

a. Prove true for $n = 1$

$$13 \times 6^1 + 2 = 100$$

which is divisible by 5

Assume true for $n = k$

$$13 \times 6^k + 2 = 5M$$

where m is a +ve integer

Prove true for $n = k + 1$

$$13 \times 6^{k+1} + 2 = 13 \times 6^k \times 6 + 2$$

$$= (13 \times 6^k + 2 - 2) \times 6 + 2$$

$$= (5M - 2) \times 6 + 2$$

$$= 30M - 10$$

$$= 5(6M - 2)$$

which is divisible by 5. If the formula is true for

$n = k$, it is true for $n = k + 1$.

Now we know it is true for $n = 1$ so it follows that it must be true for $n = 1 + 1 = 2$ and because it is true for $n = 2$ it must be true for $n = 3$ and so on for all positive integer values of n .

b.i. $x = \sin^2 5t + 2$

when $x = \frac{5}{2}$,

$$\frac{5}{2} = \sin^2 5t + 2$$

$$\sin^2 5t = \frac{1}{2}$$

$$\sin 5t = \pm \frac{1}{\sqrt{2}}$$

$$5t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \dots$$

$$t = \frac{\pi}{20}, \frac{3\pi}{20}, \frac{5\pi}{20}, \dots$$

Particle first at $x = \frac{5}{2}$

after $\frac{\pi}{20}$

ii. $x = \sin^2 5t + 2$

$$v = 10 \sin 5t \cos 5t$$

$$v = 5 \sin 10t$$

when $t = \frac{\pi}{20}$, $v = 5 \sin \pi$,

so $v = 0$

positive value means particle is travelling in the positive direction at this time.

iii. $x = \sin^2 5t + 2$

$$v = 5 \sin 10t$$

$$a = 50 \cos 10t$$

$$a = 50(1 - 2 \sin^2 5t)$$

$$a = 50 - 100 \sin^2 5t$$

now $x = \sin^2 5t + 2$

$$x - 2 = \sin^2 5t$$

$$a = 50 - 100(x - 2)$$

$$a = 250 - 100x$$

iv. $a = 50 - 100(x - 2)$

is of the form

$$a = \text{constant} - n^2(x - a^2)$$

Where $n = 10$ and $a^2 = 2$

so $a = \sqrt{2}$

v. $n = 10$

$$\text{period} = \frac{2\pi}{n}$$

$$= \frac{2\pi}{10}$$

$$= \frac{\pi}{5} \text{ seconds.}$$